

CS23633 – Cloud Computing Laboratory

Experiment 3:

Setting Up Hybrid Cloud Connectivity Using Cisco Packet Tracer (Simulation)

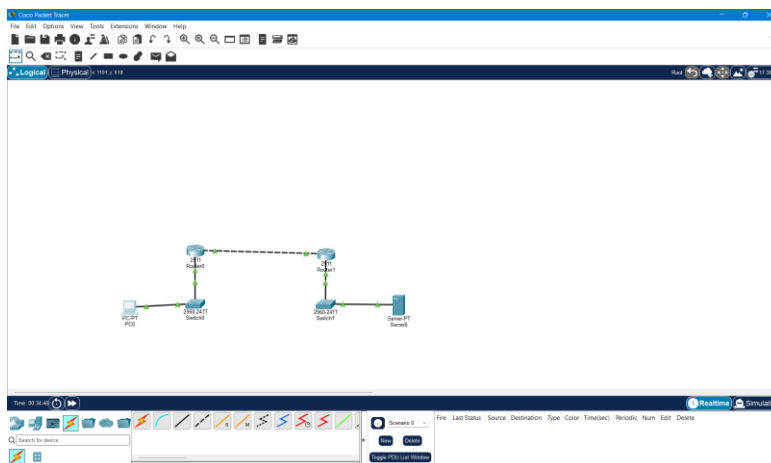
Aim:

To simulate a secure hybrid cloud connection between an on-premises data center and a cloud network using routers, switches, subnets, and routing in Cisco Packet Tracer.

Software Used:

- Cisco Packet Tracer
- Windows 11 (or your OS)

Network Topology:



Hybrid cloud network topology.

- PC
- Switch0
- Router0
- Router1
- Switch1
- Server
- All green links

Router0 Configuration

```

Router0
Physical Config CLI Attributes
IOS Command Line Interface

Router(config-if)#
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet0/1
Router(config-if)#ip address 10.10.10.1 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#do show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  10.10.10.1     YES manual up          down
GigabitEthernet0/2  unassigned     YES unset  administratively down down
Vlan1          unassigned     YES unset  administratively down down
Router(config)#
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up

Router#show ip interface brief
Interface      IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0  192.168.1.1    YES manual up          up
GigabitEthernet0/1  10.10.10.1     YES manual up          up
GigabitEthernet0/2  unassigned     YES unset  administratively down down
Vlan1          unassigned     YES unset  administratively down down
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 10.0.0.0 255.255.255.0 10.10.10.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#

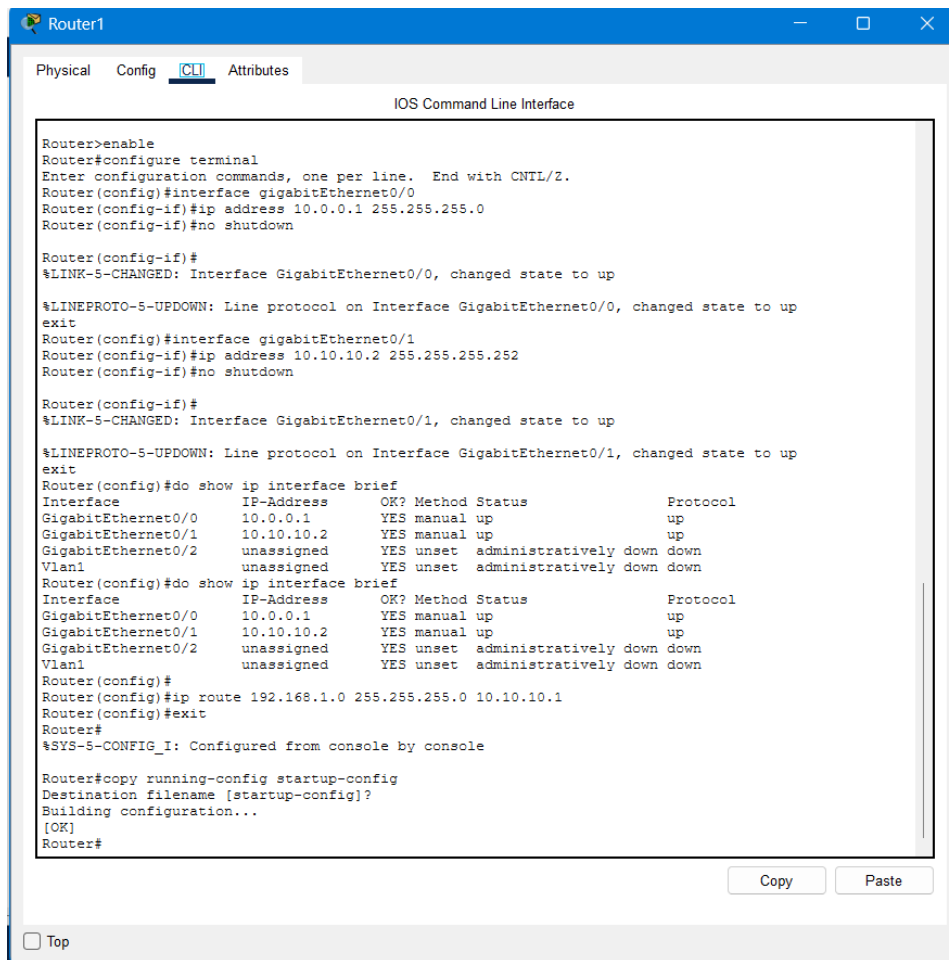
```

Copy Paste

☐ Top

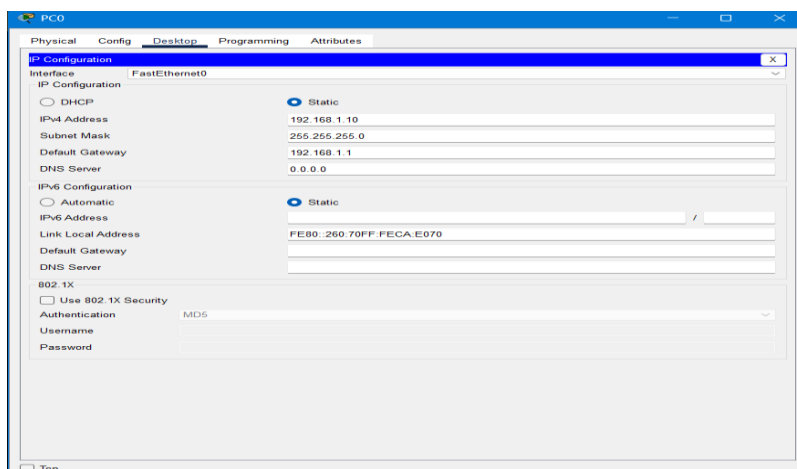
Router0 interface configuration.

Router1 Configuration:



Router1 interface configuration.

PC Configuration:



PC IP configuration.

Server Configuration:

The screenshot shows the 'Server0' configuration window with the 'Config' tab selected. The 'IP Configuration' section is expanded, showing the following settings:

- IP Configuration:**
 - ☐ DHCP
 - ☒ Static
 - IPv4 Address: 10.0.0.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 10.0.0.1
 - DNS Server: 0.0.0.0
- IPv6 Configuration:**
 - ☐ Automatic
 - ☒ Static
 - IPv6 Address: (empty)
 - Link Local Address: FE80: 260:5CFF:FEC8:39B8
 - Default Gateway: (empty)
 - DNS Server: (empty)
- 802.1X:**
 - ☐ Use 802.1X Security
 - Authentication: MDS
 - Username: (empty)
 - Password: (empty)

At the bottom, there is a 'Top' button.

Server IP configuration.

Static Route Configuration:

The image shows two side-by-side screenshots of the 'Router0' and 'Router1' configuration windows, both with the 'CLI' tab selected.

Router0 CLI:

```

Router con0 is now available

Press RETURN to get started.

Router>
Translating "\*"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 3 subnets, 3 masks
S    10.0.0.0/24 [1/0] via 10.10.10.2
C    10.10.10.0/30 is directly connected, GigabitEthernet0/1
L    10.10.10.1/32 is directly connected, GigabitEthernet0/1
C    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
L    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0

Router>
  
```

Router1 CLI:

```

*LINE-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
*LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Router(config)#do show ip interface brief
Interface  IP-Address  OK? Method Status  Protocol
GigabitEthernet0/0  10.0.0.1    YES manual up        up
GigabitEthernet0/1  10.10.10.2  YES manual up        up
GigabitEthernet0/2  unassigned  YES unset  administratively down down
Vlan1        unassigned  YES unset  administratively down down
Router(config)#do show ip interface brief
Interface  IP-Address  OK? Method Status  Protocol
GigabitEthernet0/0  10.0.0.1    YES manual up        up
GigabitEthernet0/1  10.10.10.2  YES manual up        up
GigabitEthernet0/2  unassigned  YES unset  administratively down down
Vlan1        unassigned  YES unset  administratively down down
Router(config)#
Router(config)#ip route 192.168.1.0 255.255.255.0 10.10.10.1
Router(config)#exit
Router#
*VRF-5-CONFIG I: Configured from console by console
Router#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

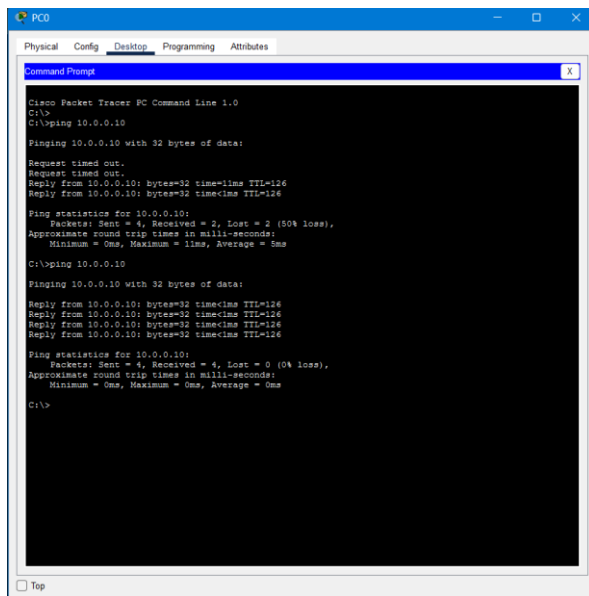
Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 4 subnets, 3 masks
C    10.0.0.0/24 is directly connected, GigabitEthernet0/0
L    10.0.0.1/32 is directly connected, GigabitEthernet0/0
C    10.10.10.0/30 is directly connected, GigabitEthernet0/1
L    10.10.10.2/32 is directly connected, GigabitEthernet0/1
S    192.168.1.0/24 [1/0] via 10.10.10.1

Router#
  
```

Static routing configuration.

Connectivity Test:



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 5ms
C:\>ping 10.0.0.10
Pinging 10.0.0.10 with 32 bytes of data:
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Reply from 10.0.0.10: bytes=32 time=1ms TTL=126
Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```

Successful ping from the PC to the cloud server.

Result:

The hybrid cloud network was successfully simulated using Cisco Packet Tracer. Two routers were configured with IP addressing and static routing to interconnect the on-premises network and the cloud network. End-to-end communication between the PC and the server was verified successfully using the ping command, confirming successful hybrid cloud connectivity.